SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1– Secure Infrastructure Architecture

Course Code:	CSA5803	**Course Title:**	DEVSECOPSENGINEERING AND APPLICATION SECURITY
CO Assessed:	CO4 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1
Weightage:	30% of CIA	**Total Marks:**	100
CO 4 Statement:	Design Secure Infrastructure Architecture using DiagrammeR or Drawio.		
Student Name:	Lokesh kumar v	**Reg. No.:**	192521170
Section / Batch:	Sloct c	**Date:**	18/8/26

Instructions to Students

- Design a Secure Infrastructure Architecture for the selected application/system using DiagrammeR or Draw.io.
- Include essential components such as Firewall, WAF, Load Balancer, Web/Application Server, Database, IAM, IDS/IPS, and Monitoring/Logging wherever applicable.
- Clearly represent network zones, data flow, security controls, and trust boundaries in the architecture.
- Ensure the diagram is original, professionally formatted, clearly labelled, and submitted along with its editable source file.

Question Type	Focus Area	Questions	Marks
Secure Infrastructure Architecture	Conceptual Understanding	Q1	Q1 –100
	GRAND TOTAL:	100 Marks	

Code of Conduct

I lokesh kumar v(1925211070) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action

lokesh kumar.

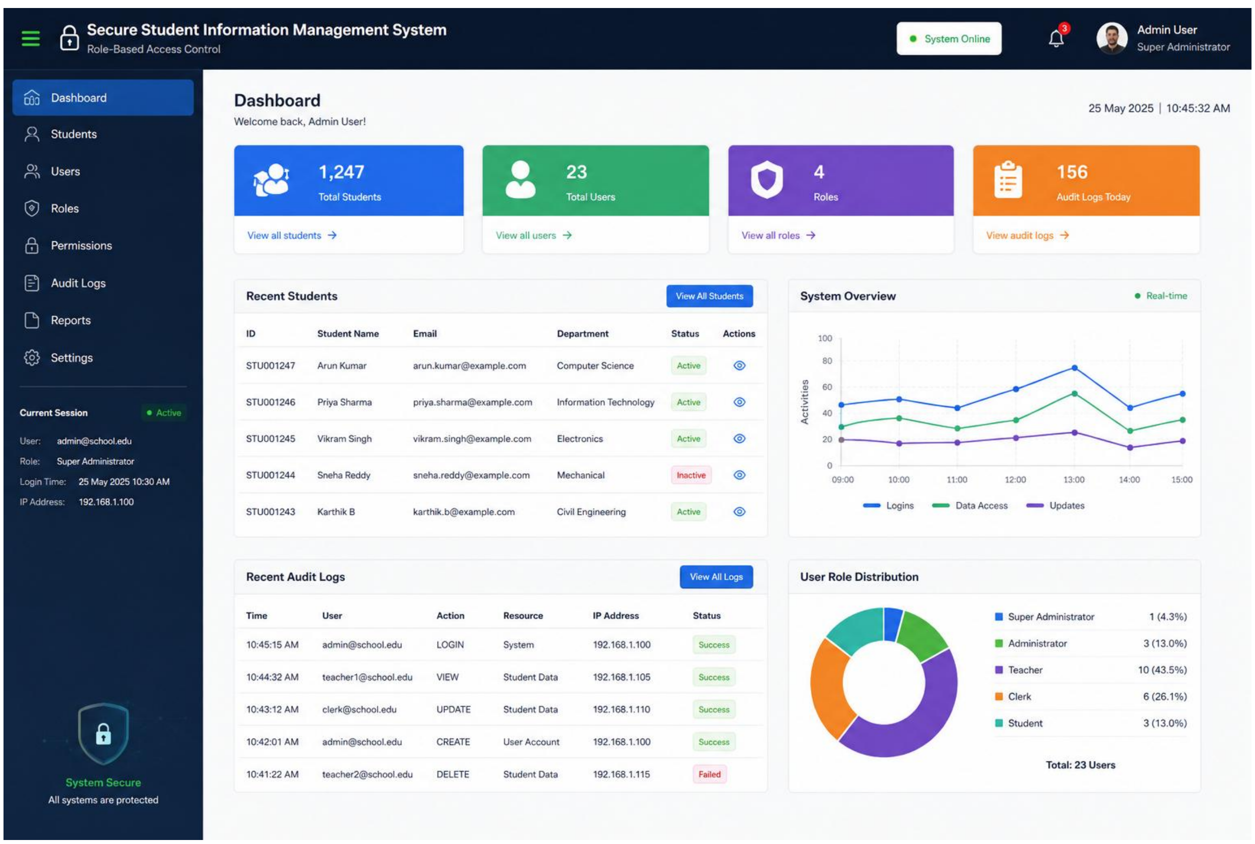
Signature of the Student:

Secure College Campus Network Architecture

1. Aim

To design a **secure college campus network** that provides controlled and isolated access for students, faculty, laboratories, administration, servers, and guest users using **VLANs, firewalls, NAC, IDS/IPS, and centralized logging**.

2. Proposed Architecture – 8 Marks



3. VLAN Design

,AN	twork	ers/Devices	rpose
,AN 10	ıdent	ıdent PCs & Wi-Fi	ademic access
,AN 20	culty	culty systems	culty resources
,AN 30	boratory	b computers	ctical/lab access
,AN 40	rver	b, DB, DNS, File servers	tical resources
,AN 50	est	est Wi-Fi	ernet only
,AN 60	ministration	fice/Admin PCs	nsitive operations
,AN 70	ınagement	itches, APs, security devices	twork management

4. Security Mechanisms

Firewall

- Filters incoming and outgoing traffic.
- Controls communication between VLANs.
- Blocks unauthorized access.
- Provides NAT and VPN protection.

NAC

- Uses **802.1X/RADIUS** authentication.
- Allows only authorized devices.
- Assigns users to the correct VLAN.
- Places unknown devices in a quarantine network.

IDS/IPS

- **IDS** detects suspicious network activity.
- **IPS** automatically blocks malicious traffic.
- Monitors important network segments.
- Generates security alerts.

Centralized Logging

- Collects logs from firewall, servers, switches, NAC and IDS/IPS.
- Stores security events centrally.
- Helps detect abnormal activity.
- Supports auditing and incident investigation.

5. Access-Control Policy

urce	**stination**	**cess**
ıdent VLAN	ernet	Allow
ıdent VLAN	rver VLAN	nited
ıdent VLAN	min VLAN	Deny
culty VLAN	ademic Servers	Allow
ɔ VLAN	quired Servers	Allow
est VLAN	ernet	Allow

urce	stination	cess
est VLAN	ernal Network	Deny
min VLAN	rver VLAN	Allow
nagement VLAN	twork Devices	Allow

6. Draw.io Design Steps

In **Draw.io/diagrams.net**:

1. Add **Internet → Router → Firewall → IDS/IPS → Core Switch**.
2. Connect the Core Switch to the **access switches**.
3. Create separate VLAN boxes for **Student, Faculty, Lab, Server, Guest and Administration**.
4. Add **NAC/RADIUS** for authentication.
5. Add **Wi-Fi Access Points** for student, faculty and guest wireless networks.
6. Add a **SIEM/Centralized Logging Server** connected to security devices.
7. Label every VLAN and show blocked/allowed connections.

8. Security Flow

Device → NAC Authentication → VLAN Assignment → Firewall/ACL → IDS/IPS Inspection → Authorized Resource → Centralized Logging

9 Result & Conclusion

The proposed architecture provides **secure VLAN segmentation, controlled access, device authentication, intrusion detection, guest isolation and centralized monitoring**. Sensitive administration and server resources are protected from unauthorized student, laboratory and guest access.